



Maheshchandra Hegde

Founder & Technology Leader at Technoayan Digital Transformation Services Pvt. Ltd. | Building AI-driven 3D asset platforms and 3d computing solutions @srushtilabs.com

Innovative technologist and product strategist with 18+ years of experience architecting end-to-end digital products, interactive 3D experiences, and domain-specific SaaS platforms. Co-Founder and Technology Leader at Technoyana Digital Transformation Services Pvt. Ltd., driving product architecture, cloud scalability, and creative technical vision.

Expertise spans modern web/mobile application stacks (React, TypeScript, Angular, Web Components, Node.js, Firebase/GCP, AWS), interactive 3D/PBR pipelines, real-time spatial computing, and design-to-code automation across both high-velocity startups and mission-critical enterprise systems.

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Technoyana & SrushtiLabs: Generative 3D Platforms & Spatial Computing

Voxelforge AI

Modular 3D Generative AI Platform | srushtilabs.com/voxelforge/

- **Role & Entity:** Founder & CTO at SrushtiLabs (Technoyana Digital Transformation Services Pvt. Ltd.).
- **Core Product:** Developed Voxelforge AI, a generative AI product creating modular 3D assets based on user prompt inputs.
- **Gaming & Visualization:** Produces game-ready, low-poly modular 3D models and bundles optimized for immediate integration into Unreal Engine, Unity, and real-time WebGL engines.
- **Prototyping Speed:** Automates prompt-to-3D geometry conversion, texture baking, and polygon reduction, cutting 3D asset turnaround from days to minutes.
- **Live Application:** Fully deployed and operational at srushtilabs.com/voxelforge/.

Ayam3d (Ayam = Dimension)

Parametric AI 3D Model Generation | ayam3d.in

- **Concept & Vision:** Extended the Voxelforge concept into Ayam3d ('Ayam' meaning Dimension), focusing on parametric-based 3D model generation from natural language prompts.
- **Instant Generation:** Emphasizes generating structured 3D models in seconds utilizing custom trained AI models and geometric constraint solvers.
- **Investor Interest:** Successfully generated early investor interest, validating the market demand for real-time generative 3D synthesis.
- **Active Development:** Currently under active development; actively seeking partners and promoters to help scale the AI pipeline.
- **Application Reference:** Preliminary concept documentation available at ayam3d.in.

Cyient / PHILIPS: Intellispace Critical Care & Anesthesia (ICCA)

Philips ICCA

Intellispace Critical Care & Anesthesia

- **Role:** Senior Technology Leader (React / NodeJS)
- **Tenure:** Nov 2023 – Mar 2025
- **Client & Facility:** Cyient Limited for Philips Healthcare, Bangalore
- **Operational Reality:** Mission-critical hospital ICU and anesthesia suites operating continuously 24/7 with zero margin for error.
- **Delivery Mandate:** Delivered clinical software passing all stringent quality, reliability, and security gates established by Philips and hospital networks.

1. Rigorous Quality Benchmarks & Zero-Downtime Reliability

Successfully delivered the software to meet demanding medical quality requirements set by Philips and healthcare providers. Architected fail-safe clinical data rendering with zero memory leaks, ensuring bedside patient monitoring displays remain completely responsive across multi-day continuous operations.

2. Comprehensive Security Architecture & Policy Enforcement

Enforced strict enterprise cybersecurity measures as per Philips global policies: integrated Multi-Factor Authentication (MFA), protected against Cross-Site Scripting (XSS) and injection vulnerabilities, enforced role-based access control (RBAC), and maintained immutable audit logs for sensitive patient medical data.

3. End-to-End Best Practices & Scalable Modular Delivery

Led the end-to-end technical execution—from architectural design and modular component breakdown to automated test suites and production deployment. Delivered a fully supported software product backed by complete documentation and service support protocols.

Ness Digital Engineering & Moonraft Innovation Labs

Ness Digital Engineering

Senior Analyst / Tech Lead | Oct 2019 – Apr 2020

- **Project Focus:** Entertainment Management Application for a major client.
- **Technology Stack:** Angular, TypeScript, AWS cloud infrastructure, Node.js.
- **System Architecture:** Architected the complete single-page application structure, establishing clean state management and secure API communication layers.
- **Standards & Best Practices:** Enforced accessibility standards (WCAG), code quality linters, and strict frontend security guidelines across the engineering team.
- **Technical Leadership:** Led the development squad through agile sprints, code reviews, and end-to-end technical deployment to client environments on schedule.

Moonraft Innovation Labs

UI Architect | May 2019 – Jul 2019 (UST Global Specialized Unit)

- **Enterprise Design System:** Architected and engineered a centralized cross-framework UI Design System using LitElement Web Components, React, AngularJS, Node.js, and npm packaging.
- **Eliminated Duplication:** Stored all standardized UI components in a central repository, preventing teams from reinventing the wheel for new client projects.
- **Unified Look & Feel:** Guaranteed visual consistency and brand coherence across all digital products from a single, shared codebase.
- **Automated Upgradability:** When the core component library is updated, client projects automatically inherit updates and enhancements via standard package management.
- **Luxury Hotel Mobile App:** Led mobile application development for a premier Indian hotel group (TLC Group of Hotels) built using Ionic and Angular.

UST Global / CISCO: Stadium Vision Director Migration

Cisco Stadium Vision

Stadium Vision Director Enterprise Platform

- **Role:** Senior Analyst & Associate Project Manager
- **Tenure:** Feb 2015 – Feb 2019 (4+ Years Dedicated Tenure)
- **Client:** Cisco Systems India Private Limited via UST Global
- **Product Scope:** Centralized digital media, live video distribution, and dynamic stadium display management deployed at major international sporting arenas.
- **Core Mandate:** Modernize a massive legacy application suite with numerous active sub-applications and active global customers.

1. Phased Zero-Downtime Migration from Legacy Flash/Flex

The platform was originally built using Adobe Flash, Flex, and ActionScript. Took technical leadership in architecting and executing a phased migration to modern Angular and React modules without interrupting active stadium operations or breaking existing client workflows.

2. Hybrid Interoperability & Multi-Stack Engineering

Maintained seamless interoperability between legacy components and modernized Single Page Application (SPA) modules using TypeScript, Angular, React, and Node.js backend services until complete deprecation was achieved.

3. Sub-Application Management & Sustained Excellence

Governed multiple parallel UI sub-applications serving venue operators, technicians, and broadcast controllers. Awarded multiple Certificates of Excellence by UST Global (2015, 2016, 2018) for client-first commitment and outstanding delivery.

ThoughtFocus Technologies: Aftermarket Parts 3D Explorer & Enterprise Solutions

Aftermarket Parts Explorer

UI Architect & Tech Lead | Oct 2011 – Mar 2014

- **Platform Scope:** Massive parts marketplace serving aftermarket automotive & industrial parts vendors and buyers.
- **Scale & Catalog:** Supported millions of parts across hundreds of manufacturer catalogs with deep search capabilities.
- **Interactive 3D Viewer:** Architected an interactive 3D interface allowing buyers to inspect 3D models and dimensions of thousands of components before purchase.
- **Full E-Commerce Flow:** Integrated shopping cart, quotation workflows, and secure payment gateway integrations.
- **Technologies Used:** Adobe Flex, ActionScript, Mate architectural framework, HTML, CSS, and backend data services.

Enterprise UI Solutions

Prototyping & Multi-Client Solutions Delivery

- **Design-to-Code Pipeline:** Established an early user involvement process—validating concepts with interactive mockups and prototypes prior to delivery.
- **Enterprise Search App:** Guided team to design and bootstrap a high-speed enterprise search application with advanced filtering and metadata indexing.
- **Dairy / Milk Corporation App:** Led the architecture and delivery of an operational management application for a large milk corporation handling distribution and supply logistics.
- **Technology Stack:** HTML, CSS, JavaScript, TypeScript, Adobe Flex, Flash, and early Single-Page Application patterns.
- **Mentorship & Guidance:** Coached junior engineers on clean modular UI architecture and user-centered design standards.

Cisco & Vodafone BMS Dashboards & Technoyana Specialized Engineering

Cisco & Vodafone Dashboards

Building Management Systems (BMS) Unified Interface

- **Facility Dashboard at Cisco:** Designed and developed an interactive facility dashboard at Cisco Systems campus as a full-time consultant (2009–2010).
- **Unified BMS Control:** Connected to and visualized diverse Building Management Systems: Air Handling Units (AHUs), Variable Refrigerant Volume (VRVs), industrial chillers, and power meters.
- **Environmental Telemetry:** Provided facility managers with real-time operational status, thermal mapping, energy consumption, and alarm thresholds.
- **Formal Cisco Recognition:** Received official Certificate of Appreciation from Cisco leadership in 2010 for outstanding visualization and operational utility.
- **Vodafone Solutions:** Delivered specialized data visualization dashboards for Vodafone operations to monitor network and facility metrics.

Specialized Engineering Projects

CAD/CAM, CNC, 3D Visualization & Mobile Products

- **Architectural Visualization:** Delivered multiple architectural 3D visualizations, spatial renderings, and walkthroughs for commercial and real estate projects.
- **CAD/CAM & CNC Development:** Led customized CNC machine development and automated CAD/CAM toolpath pipelines, bridging physical fabrication with digital software.
- **SellAny Mobile Marketplace:** Built and launched 'SellAny'—a consumer mobile application allowing users to list and sell items effortlessly.
- **Payment Parking Mobile App:** Led an engineering team to develop an automated parking payment mobile application with real-time slot occupancy tracking.

CAE Simulation Technologies: Flight Simulation Visual Databases

CAE Flight Simulation Systems

Visual Database Developer | Sep 2007 – Apr 2008

- **Company Profile:** CAE (Canadian Aeronautical Engineering) — world leader in high-fidelity civil aviation and military aircraft flight simulation training devices.
- **Visual Database Engineering:** Focused on the design, modeling, and generation of comprehensive visual databases that power real-time full flight simulators (FFS).
- **Synthetic World Creation:** Constructed accurate, high-fidelity 3D synthetic environments including runways, airport terminals, terrain elevation, navigation lighting, and approach corridors.
- **Structural & Coordinate Precision:** Fed precise 3D visual, topographical, and structural collision details into databases calibrated to real-world GIS coordinates.
- **Civil & Military Applications:** Supported pilot training simulators for major commercial airliners as well as defense tactical training aircraft.

3D Modeling, Texturing & Performance

Photorealistic Texture Generation & Locked 60 FPS Budgets

- **3D Modeling & Environment:** Created detailed 3D models of aircraft structures, ground support vehicles, building structures, and regional geographic landmarks.
- **Photoshop Texture Pipelines:** Generated realistic multi-layer textures using Adobe Photoshop, including satellite imagery alignment, seasonal variants, and night-vision/illumination maps.
- **Locked 60 FPS Real-Time Budgets:** Engineered all 3D assets under strict polygon limits and Level of Detail (LOD) hierarchies to guarantee zero-latency 60 FPS simulator performance.
- **Pure Visual & Design Craft:** Specialized design role bridging artistic 3D visualization, photorealistic texturing, and rigorous database integration.
- **Foundational 3D Discipline:** This early career experience in simulation systems established the deep graphics and spatial computing expertise that later led to Voxelforge AI and Ayam3d.

Comprehensive Skillset Matrix Across 18+ Years

1. Scalable Frontend & Modular Architecture

- React, Angular, TypeScript, JavaScript, Node.js, Web Components (LitElement).
- Architected Enterprise UI Design Systems used across diverse development teams.
- State management, micro-frontends, REST, WebSockets, and streaming telemetry.
- 4-year zero-downtime migration of Cisco Stadium Vision from Flash/Flex to Angular/React.

3. High Reliability & Operational Security

- Delivered Philips ICU healthcare software meeting stringent clinical safety benchmarks.
- Strict cybersecurity enforcement: Multi-factor authentication (MFA), XSS mitigation, RBAC.
- Offline-first mobile architecture (IndexedDB/Service Workers) with conflict-free cloud sync.
- Building Management System (BMS) unified interface for AHUs, chillers, and telemetry (Cisco).

2. Spatial 3D, CAD & Generative AI

- Voxelforge AI (generative modular 3D for Unreal/Unity/WebGL) & Ayam3d (parametric 3D).
- Architectural 3D visualization, animated walkthroughs, and CAD/CAM CNC machine development.
- Flight simulator 3D terrain and aircraft databases under locked 60 FPS budgets (CAE).
- Interactive 3D parts inspection explorer supporting millions of catalog components.

4. Agile Leadership & Startup Delivery

- Founder/Co-Founder: Technoyana, Twitan.com, SrushtiLabs, InnoBrik.
- Associate Project Manager at UST Global; managed cross-functional squads for Cisco.
- Early user prototyping, design-to-code pipelines, and active stakeholder governance.
- Honored with Cisco Appreciation (2010) and 3 UST Global Excellence Awards (2015, 2016, 2018).

Major Achievements & Milestones

1. Zero-Downtime Global Stadium Platform Migration

- **Platform Scope:** Modernized Cisco Stadium Vision Director deployed at international sporting arenas worldwide across a 4-year dedicated tenure.
- **Engineering Mandate:** Led the phased migration from legacy Adobe Flash/Flex to modern Angular and React without a single minute of venue downtime.
- **Operational Excellence:** Maintained live broadcast reliability and client workflows across dozens of active sub-applications.
- **Industry Honors:** Awarded 3 consecutive UST Global Excellence Awards (2015, 2016, 2018) for exceptional delivery and client commitment.

2. Healthcare-Grade ICU Software Delivery (Philips)

- **Clinical Mandate:** Delivered web-tier software for Philips Intellispace Critical Care & Anesthesia (ICCA) operating in 24/7 ICU suites.
- **Stringent Standards:** Passed demanding hospital quality, safety, and zero-leak reliability benchmarks for mission-critical bedside monitoring.
- **Enterprise Security:** Enforced Multi-Factor Authentication (MFA), role-based access control (RBAC), and strict patient data protection policies.

3. Generative 3D Asset & Spatial AI Commercialization

- **Product Innovation:** Founded and launched Voxelforge AI (srushtilabs.com/voxelforge/), generating modular 3D assets from natural language.
- **Game-Ready Bundles:** Automated geometry synthesis, texture baking, and polygon reduction for instant Unreal Engine, Unity, and WebGL integration.
- **Parametric AI (Ayam3d):** Developed Ayam3d (ayam3d.in) for parametric 3D generation in seconds, earning early investor interest and partner validation.

4. Enterprise UI Design System & BMS Facility Control

- **Design System (Moonraft):** Engineered centralized Web Components design system (LitElement, React, Angular) eliminating duplicated effort across project teams.
- **Cisco Facility Dashboard:** Delivered unified BMS dashboard for AHUs, chillers, and power telemetry; awarded Cisco Certificate of Appreciation (2010).
- **Aftermarket 3D Explorer:** Architected interactive 3D parts explorer at ThoughtFocus supporting millions of catalog components and e-commerce flows.

Challenges Faced & Resolutions Implemented

Challenge 1: Zero-Downtime Migration of Global Legacy Platform

- **The Challenge:** Cisco Stadium Vision Director was built on deprecated Adobe Flash/Flex with extensive business logic serving global arenas. Upgrading risked operational downtime and customer disruption.
- **Resolution Implemented:** Architected a hybrid micro-frontend bridge allowing new Angular and React modules to communicate with legacy Flex components via a shared event bus. Migrated sub-apps incrementally with zero downtime.

Challenge 2: 24/7 Clinical Bedside Memory Stability & Security

- **The Challenge:** Philips ICU software operates non-stop for days. Memory leaks or UI freezes in bedside patient monitoring could lead to catastrophic clinical outcomes, amidst strict HIPAA/cybersecurity rules.
- **Resolution Implemented:** Enforced strict browser profiling, deterministic lifecycle cleanup on unmount, virtualized rendering for dense telemetry streams, and institutionalized MFA/RBAC security policies.

Challenge 4: Generative 3D Mesh Complexity & Performance Lag

- **The Challenge:** Different project teams at Moonraft / UST were independently building common UI widgets across React and Angular, leading to visual inconsistency, duplicated effort, and slower client delivery.
- **Resolution Implemented:** Architected an enterprise-wide UI Design System using LitElement Web Components packaged via npm. Teams across any frontend framework seamlessly consumed identical, standardized components from one repo.
- **The Challenge:** Raw AI-generated 3D meshes produced disorganized vertex topology and massive polygon counts that crashed WebGL browsers and caused severe frame drops in gaming engines.
- **Resolution Implemented:** Engineered automated post-processing pipelines for polygon decimation, procedural retopology, UV unwrapping, and texture baking—generating optimized, lightweight 3D models instantly.

Lessons Learned & Best Practices

1. Incremental Modernization Outperforms 'Big-Bang' Rewrites

- **Lesson Learned:** Complete system rewrites carry high risk, long delay cycles, and potential customer rejection. Incremental migration allows continuous production value delivery.
- **Best Practice:** Use strangler-fig patterns, modular micro-frontends, and backward-compatible APIs to modernize enterprise platforms step-by-step while maintaining uninterrupted business operations.

2. Security & Compliance Must Be Architected from Day Zero

- **Lesson Learned:** Retrofitting authentication, RBAC, encryption, and audit trails into an established product is costly, disruptive, and prone to severe security vulnerabilities.
- **Best Practice:** Embed security controls (MFA, automated dependency scanning, sanitization, role matrices) into the CI/CD pipeline from inception as standard enterprise non-negotiables.

3. Centralized Design Systems Deliver Compounding ROI

- **Lesson Learned:** Without shared component governance, engineering squads create fragmented code, brand inconsistency, and duplicate maintenance burdens across projects.
- **Best Practice:** Treat internal design systems as first-class products with semantic versioning, comprehensive documentation, and framework-agnostic standards (Web Components).

4. Early Interactive Prototyping Resolves Requirement Risk

- **Lesson Learned:** Specification documents alone fail to capture operational nuances; misunderstandings surface late during production deployment when changes are 10x more costly.
- **Best Practice:** Involve stakeholders and users early using clickable, interactive prototypes to validate workflows, usability, and data contracts before committing heavy backend engineering.

Suggestions for Future Process Improvements

1. Automated IT Governance & Self-Service Developer Portals

- **Objective:** Establish standardized internal developer platforms (IDPs) and automated infrastructure-as-code (IaC) templates for rapid, secure project spin-ups.
- **Expected Benefit:** Cuts new system bootstrapping and environment provisioning time from weeks to hours while ensuring automated compliance with corporate security baselines.

2. AI-Assisted Engineering Toolchains & Code Quality Gates

- **Objective:** Integrate generative AI tooling into enterprise workflows for automated unit test generation, synthetic data simulation, and design-to-code asset generation.
- **Expected Benefit:** Boosts developer productivity by 30–40%, accelerates testing cycles, reduces boilerplate coding, and frees engineering teams to focus on strategic core systems.

3. Unified Enterprise Telemetry & Proactive Observability

- **Objective:** Synthesize application performance monitoring (APM), network health, and facility sensor telemetry (BMS) into unified operational command dashboards.
- **Expected Benefit:** Transforms IT operations from reactive issue response to predictive anomaly detection, preventing service degradation and minimizing system downtime.

4. Structured Knowledge Sharing & Continuous Upskilling

- **Objective:** Institute regular cross-functional architecture reviews, tech talk forums, and sponsored enterprise certification tracks (Cloud, Security, ITIL, Agile).
- **Expected Benefit:** Breaks down engineering silos, accelerates cross-pollination between teams, and ensures enterprise alignment with rapidly evolving modern industry standards.

Summary

- ✓ **Design and Engineering Synthesis:** Holding a postgraduate degree in Human Interface Design (M.S. in Computing, UK) combined with an engineering degree in Electronics and Communication (B.E.), I bring a rare, balanced capability spanning both human-centered UX design and deep technical software architecture.
- ✓ **Proven Breadth Across Startup Velocity & Enterprise Scale:** My 18+ years of experience encompass leading agile, high-velocity startup product incubation (Technoyana, SrushtiLabs, Twitan, InnoBrik) as well as managing complex multi-year enterprise projects with stringent quality, security, and uptime mandates (Philips ICU Healthcare, Cisco Systems, UST Global, Ness).
- ✓ **Handling Products Across Every Stage of the Lifecycle:** Demonstrated expertise across every product phase: early discovery prototypes, building reusable design systems, legacy system migrations (Flash to Angular/React), production deployment, security compliance, and long-term customer support.
- ✓ **Pledge for Self-Training & Certification:** I am fully committed to the success of the organization and team. As per the specific operational requirements of the Senior Manager – IT role, I pledge to proactively upskill myself and complete any required enterprise certifications (e.g., Cloud Architecture, ITIL, Cybersecurity, or project governance) to align completely with team goals and project needs.

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